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Applied Information Extraction in Python

Today's goal • 3/3 items

Module 1
Information Extraction

Module 2
Named Entity Recognition (NER)

Module 3
Neural Network Models

Module 4
Transformers Transform Information Extraction

Language Models

Language Models (LMs)
Video • 11 min

Large Language Models (LLMs)
Video • 7 min

Knowledge Check: Language Models
Practice Assignment • Grade: 100%

What Are Transformers?

Transformers
Video • 9 min

Recent Advances in GPTs
Reading • 10 min

Knowledge Check: What Are Transformers?
Practice Assignment • 10 min

How to Build Information Extraction Models Using Transformers

Building an Information Extraction Pipeline with Transformers and LLMs
Reading • 10 min

Design an Information Extraction System for Sports News Reports
Dialogue • 15 min

Module Wrap-Up & Assignment

Assignment 4 Introduction: Build an Information Extraction Pipeline using Transformers
Video • 4 min

Build an Information Extraction Pipeline using Transformers
Programming Assignment • 3h

Module 4 Assignment
Graded Assignment • 30 min

Course Wrap Up

Course Wrap-Up
Video • 4 min

Continue Your Journey and Earn a Master of Applied Data Science Degree Online
Reading • 10 min

Course Post-Survey
Reading • 10 min

Recent Advances in GPTs

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In just a few years, **transformer-based language models** have evolved from academic experiments to tools that power entire industries. What began with modest architectures and a few hundred million parameters has grown into models with **billions, even trillions, of weights**, enabling capabilities once thought impossible.

In this reading, we'll explore:

- The evolution of major GPT and transformer models
- Key technical innovations across generations
- Real-world applications across domains
- What's coming next in this fast-moving field

From ELMo to GPT-4: The Model Timeline

Here's a chronological overview of key transformer and GPT-style models:

Model	Release	Parameters	Notes
ELMo	Early 2018	94M	Contextual word embeddings; LSTM-based
GPT-1	Mid 2018	110M	First generative transformer; unidirectional
BERT	Late 2018	340M	Bidirectional encoder; masked word prediction
GPT-2	Early 2019	1.5B	Fluent text generation at scale
RoBERTa	Mid 2019	355M	Robust BERT with dynamic masking
DistilBERT	Late 2019	66M	Lightweight BERT for edge devices
BART	Late 2019	400M	Seq2seq transformer for text generation
T5	Late 2019	11B	Unified NLP tasks under "text-to-text" format
GPT-3	Mid 2020	175B	Emergence of zero- and few-shot learning
ChatGPT (GPT-3.5)	Late 2022	~175B	Instruction-tuned; interactive dialogue
GPT-4	March 2023	(undisclosed; ~1.8T speculated)	Multimodal, more robust across tasks
Claude 3 (Anthropic)	March 2024	-	Strong performance on reasoning tasks
Gemini 1.5 (Google DeepMind)	Early 2024	-	Multi-modal with long-context memory
LLaMA 2 (Meta)	Mid 2023	7B–70B	Open weights; strong fine-tuning options
Mistral 7B / Mixtral	Late 2023	7B–12.9B (MoE)	Fast open models with sparse routing
Command R / xAI Grok	2024–2025	-	Enterprise fine-tuning and reasoning

What's Driving These Advances?

As GPT-style models scaled, several breakthroughs helped them evolve from simple text generators to general-purpose agents:

1. Instruction Tuning
    a. Makes models better at following user prompts
    b. Core to tools like ChatGPT, Gemini, and Claude
2. Emergent Abilities
    a. Some skills (e.g., math, translation) appear only at large scales
    b. Example: GPT-3 can estimate modular arithmetic; GPT-2 cannot
3. Multimodal Inputs
    a. GPT-4, Gemini, and Claude can handle **text + images**
    b. Enables visual question answering, image captioning, document understanding
4. Retrieval-Augmented Generation (RAG)
    a. Combines LLMs with external databases
    b. Used in healthcare, legal research, enterprise knowledge search
5. Sparse Mixture-of-Experts (MoE)
    a. Activates only parts of the model per input
    b. Powers models like Mixtral to boost efficiency with less computing.

Where Are They Used?

These models now power real-world systems in domains like:

Domain	Example Use Case
Healthcare	Chart summarization, SDOH extraction, patient Q&A
Legal	Contract review, e-discovery
Education	Personalized tutoring, language practice
Enterprise	Report generation, knowledge retrieval
Science	Literature review, protein structure analysis
Creative Arts	Story generation, songwriting, code autocompletion

In many cases, GPTs have matched or exceeded human-level benchmarks on reasoning, summarization, and domain-specific QA.

What's New in Modern Models?

Today's large language models are no longer just for short text generation. Recent innovations include:

- **100,000+ token contexts** (e.g., Gemini 1.5, Claude 3.5)
- **Image and audio support** (e.g., GPT-4V, Gemini)
- **Agentic abilities** (e.g., planning, browsing, code execution)

What this means for users:

- Summarizing **entire books or documents**
- Performing **multi-step tasks** autonomously
- Parsing **charts, diagrams, and PDFs**

What This Means for Information Extraction

You've built traditional IE pipelines using BiLSTMs and CRFs. Transformer models go even further:

- **End-to-end training** (no manual features)
- **Cross-domain generalization** (few-shot learning for legal/clinical/social texts)
- **Richer outputs** (e.g., "Who is the patient's primary oncologist?" → direct answer)

Your earlier work laid the groundwork. Now, these tools scale that foundation to **entire systems** that extract and reason over unstructured information.

Suggested Exploration

- **Emergent Abilities of LLMs** – Wei et al. (2022)

Reflection Prompt

You've now worked with **CRFs**, **LSTMs**, and **transformers**.

How do you think future models will balance **scale**, **efficiency**, and **control**? What risks or trade-offs might arise from giving models too much general ability?

Go to next item Completed

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